Goal Tracker: Build-Ready Spec

A web app for setting goals with deadlines, breaking them into weekly targets, logging daily progress, and seeing how far ahead or behind plan you are. Examples: marathon training, weight loss, learning a language, completing a syllabus.

1. Tech constraints

- Latest stable Flask, Python 3.11+
- No third-party packages other than Flask
- SQLite via `sqlite3` (a single local file)
- Passwords hashed with `hashlib.scrypt`; CSRF tokens from `secrets`
- Secret key read from an environment variable, with a clear error if it is missing
- Charts rendered server-side as inline SVG (no JavaScript chart library)
- Runs locally only; lives in the `lowcode-project` repo
- Mobile-first, responsive, accessible (labels, contrast, keyboard navigation)

2. Decisions made

- A user can have multiple goals, plus a combined dashboard
- Friends are mutual: both people must accept the connection
- Each goal can have several metrics, with exactly one marked primary
- On track means hitting the exact target; each goal has a user-set acceptance threshold
- Daily entries roll up into weekly totals
- Email and password login

3. Data model

Table	Key fields
`users`	id, name, email (unique), password_hash, created_at

Table	Key fields
`goals`	id, user_id, title, description (max 50 words), start_date, end_date, acceptance_pct (default 0 = exact target), share_level (private / summary / full), status
`metrics`	id, goal_id, name, unit, direction (increase / decrease), aggregation (sum / latest), start_value, final_target, is_primary
`plan_versions`	id, metric_id, version_no, end_date, final_target, reason, created_at. Version 1 is the original and is never overwritten.
`weekly_targets`	id, plan_version_id, week_start, target_value, description (max 50 words)
`entries`	id, metric_id, entry_date, value, note. One per metric per day.
`friendships`	id, requester_id, addressee_id, status (pending / accepted / declined / revoked)
`comments`	id, goal_id, author_id, body, created_at

4. Behaviour rules

- **Weekly status per metric:** a week is *met* if actual is at or past target (direction respected), or within the goal's acceptance threshold of target. Otherwise it is *behind*. Variance is shown per week and cumulatively.
- **Goal score:** driven by the primary metric. Secondary metrics show their own status chips but do not change the goal's headline status.
- **Roll-up:** cumulative metrics (e.g. miles) sum the week's daily entries; point-in-time metrics (e.g. weight) use the latest entry that week.
- **Combined dashboard:** one card per goal showing the primary metric's percent of plan, a status chip (ahead / on track / behind), weeks ahead or behind, and current streak. Units are never added together.
- **Plan generation:** weekly targets are generated from start to end date, distributed evenly, ramped linearly, or edited manually.

- **Revisions:** changing the final target or end date creates a new plan version with a required reason. The graph shows the original as a dashed line and the current plan as solid, and the history page lists every version.
- **Friends:** mutual connection only. A friend sees a goal only at the level the owner set (private, summary, or full detail). Revoking removes access immediately. Friends can comment on goals shared with them.

5. Form validation (server-side)

On any failure: re-render the same form with HTTP 400, show an error summary at the top and an inline message under each field, preserve everything the user typed, and set `aria-invalid` and `aria-describedby`. Jinja autoescape stays on.

Field	Rule
Name	Required, trimmed, max 100 characters
Email	Required, well-formed, max 254 characters, unique at registration
Password	Required, at least 10 characters
Goal title and description	Required, max 50 words
Metric name and unit	Required
Start date	Valid ISO date, not in the past
End date	Valid date, after start, at least 1 week away
Final target	Numeric, greater than 0
Acceptance threshold	Numeric, 0 to 50 (%)
Weekly target value	Numeric, greater than 0; week start within the goal's date range
Weekly target description	Required, max 50 words
Daily entry	Numeric; date within goal range and not in the future

6. Routes

Route	Purpose
`/`	Combined dashboard
`/register`, `/login`, `/logout`	Accounts
`/goals/new`	Create goal and metrics
`/goals/<id>`	Goal detail and graph
`/goals/<id>/log`	Log daily entries
`/goals/<id>/plan`	Edit weekly targets
`/goals/<id>/revise`	Change final target or end date
`/goals/<id>/history`	Version timeline
`/goals/<id>/share`	Set share level
`/friends`, `/friends/invite`	Connections
`/contact`	Optional feedback form

7. Design

Clean card layout, one accent colour, generous white space, a progress ring per goal on the dashboard, and green / amber / red chips for ahead / on track / behind. Empty states include a clear call to action. Dates display in the user's locale format.

8. Acceptance criteria

1. Any empty required field shows a visible error and keeps the other values.
2. A malformed email such as `a@b` is rejected server-side even if browser validation is bypassed.
3. Each goal has exactly one primary metric.
4. With a threshold of 0, a week counts as met only when the target is hit exactly.
5. Daily entries update the weekly roll-up and the graph immediately.
6. Revising a goal keeps the original version visible.

7. A friend never sees more than the owner's share level allows.
8. The app runs with only Flask installed.